

In other words, the satellite loses approximately 2×10^9 J.

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Using $P_{\text{input}} = \frac{m}{t}c\Delta T + h$

$$10 = \frac{m}{t}c\Delta T + h \quad \dots\dots\dots(1)$$

$$18 = \frac{3m}{t}c\Delta T + h \quad \dots\dots\dots(2)$$

Solving (1) and (2), rate of heat loss to the surrounding, $h = 6.0$ W

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